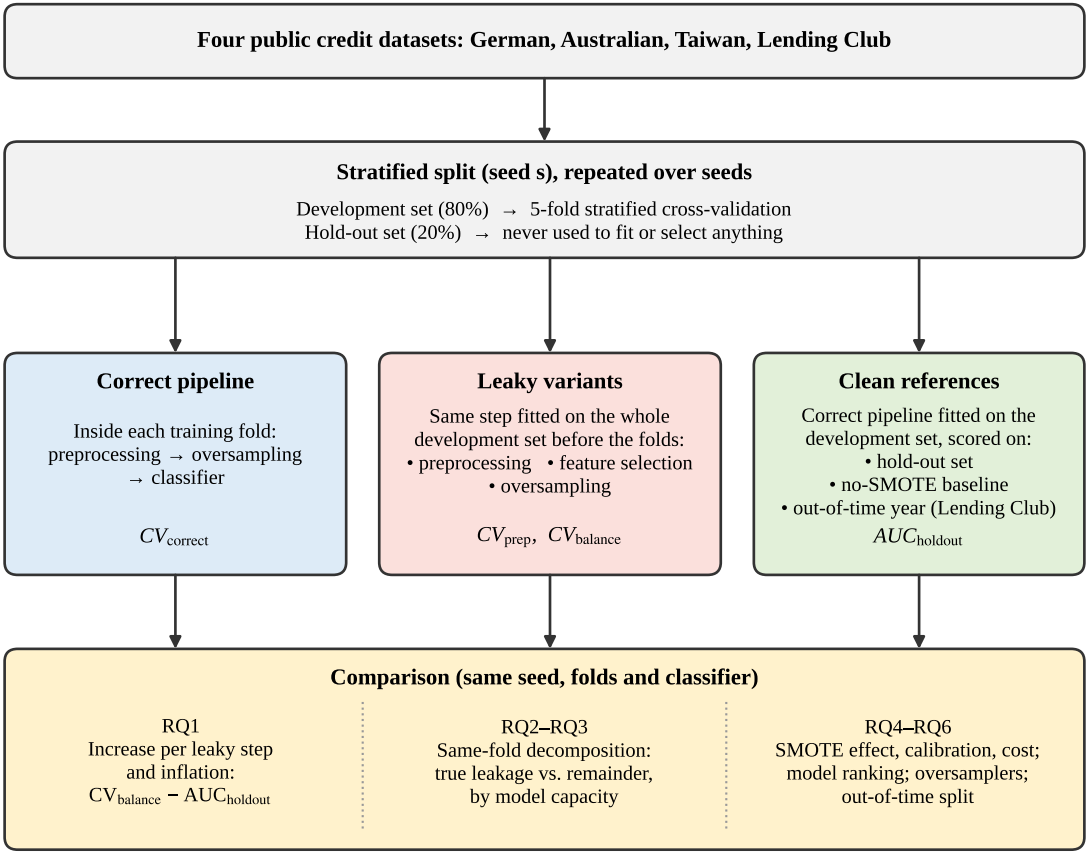
Four public credit datasets: German, Australian, Taiwan, Lending Club
Stratified split (seed s), repeated over seeds
Development set (80%) → 5-fold stratified cross-validation
Hold-out set (20%) → never used to fit or select anything
Correct pipeline
Inside each training fold:
preprocessing → oversampling
→ classifier
CV_{correct}
Leaky variants
Same step fitted on the whole
development set before the folds:
• preprocessing • feature selection
• oversampling
CV_{prep}, CV_{balance}
Clean references
Correct pipeline fitted on the
development set, scored on:
• hold-out set
• no-SMOTE baseline
• out-of-time year (Lending Club)
AUC_{holdout}
Comparison (same seed, folds and classifier)
RQ1
Increase per leaky step
and inflation:
$CV_{\text{balance}} - AUC_{\text{holdout}}$
RQ2–RQ3
Same-fold decomposition:
true leakage vs. remainder,
by model capacity
RQ4–RQ6
SMOTE effect, calibration, cost;
model ranking; oversamplers;
out-of-time split